

Mudassar Ali

Rawalpindi, Pakistan

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Summary

Mechanical Engineer with experience in manufacturing, CAD, and embedded electronics across different design, manufacturing, and robotics projects. Skilled in evaluating required product demands and innovating, manufacturing, prototyping, and designing working solutions.

Education

NUST College Of Electrical and Mechanical Engineering

Bachelor of Mechanical Engineering

Rawalpindi, Pakistan

Sep 2022 – May 2026

Experience

National Centre of Robotics and Automation (NCRA)

Intern – Robot Design & Development Lab

Rawalpindi

Jun 2025 – Jul 2025

- Contributed to the "Robodog" project, focusing on the integration of biomimetic movement and mechanical stability.
- Accelerated the design iteration cycle for structural components by leveraging rapid prototyping techniques, including 3D printing and laser cutting.
- Performed rigorous structural analysis in ANSYS and 3D modelling in SolidWorks to ensure the mechanical integrity and optimized motion of 4-bar linkage systems.

Research and Development Center (RDC), NUST

Project Contributor

Rawalpindi

Oct 2025 – Apr 2026

- Streamlined manufacturing workflows through high-precision digital documentation and mechanical design.
- Produced comprehensive 2D and 3D CAD drawings for an electric bike project, facilitating seamless fabrication transitions and reducing production errors.
- Designed and optimized a hydraulic workbench, ensuring ergonomic efficiency, load-bearing reliability, and practical applicability.

Projects

Final Year Project – VTOL UAV

Sep 2025 – May 2026

- Designed and developed a tricopter tilt-rotor VTOL drone equipped with an onboard AI detection system.
- Integrated an open-source AI vision system using a Raspberry Pi and camera to detect survivors during mass floods, covering entire regions and locating thousands of survivors; optimized planning for rescue missions and explored AI use cases in intercepting attack drones.

Skyscraper Cleaning Robot (Team Lead)

Nov 2024 – Jan 2025

- Designed an automated robotic alternative to high-risk manual labor in high-rise maintenance.
- Developed a smart wire-grip movement system utilizing tilt correction and obstacle detection to ensure operational safety and efficiency without relying on conventional, hazardous open lifts.

Autonomous Solar Cleaning Robot (Ongoing)

Apr 2024 – Present

- Assessed the feasibility of automated cleaning systems and designed a smart robot to address the flaws in current solutions.
- Integrating autonomous navigation and vision to clean multiple solar panel arrays of any layout without supervision.

Achievements & Leadership

Electric Human-Powered Vehicle Competition (EHPVC)

Team Lead

Feb 2025 – May 2025

- Led a 10-member team in the structural design, engineering, and fabrication of an electric human-powered vehicle for Pakistan's inaugural EHPV Competition.

Annual Machine Design Competition

3rd Position

2023

- Designed and manufactured a hydraulic arm recognized for excellence in mechanical conceptualization, problem-solving, and practical design execution.

Zindigi Prize Competition

Regionals (Islamabad)

2024

- Pitched a solar cleaning automation startup idea, with market penetration insights demonstrating strong entrepreneurial acumen and market awareness.

Technical Skills & Interests

Technical Skills: SolidWorks, ANSYS, Rapid Prototyping (3D Printing, Laser Cutting), CAD Design, Intermediate Programming (Python and C++), Problem Identification & Solving.

Interests: Robotics, Machine Learning, Automation, Innovation, Product Design and Execution.